

Coordination of Care for Patients with Gastrointestinal Diseases

OMFS 512: Biomedical and Clinical Integrated Sciences

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Review medical conditions associated with the GI tract.
2. Discuss the oral manifestations associated with certain GI diseases.
3. Review dental treatment and modifications.

Competency statements met

1.3, 1.5, and 2.1

Gastroesophageal Reflux Disease (GERD)

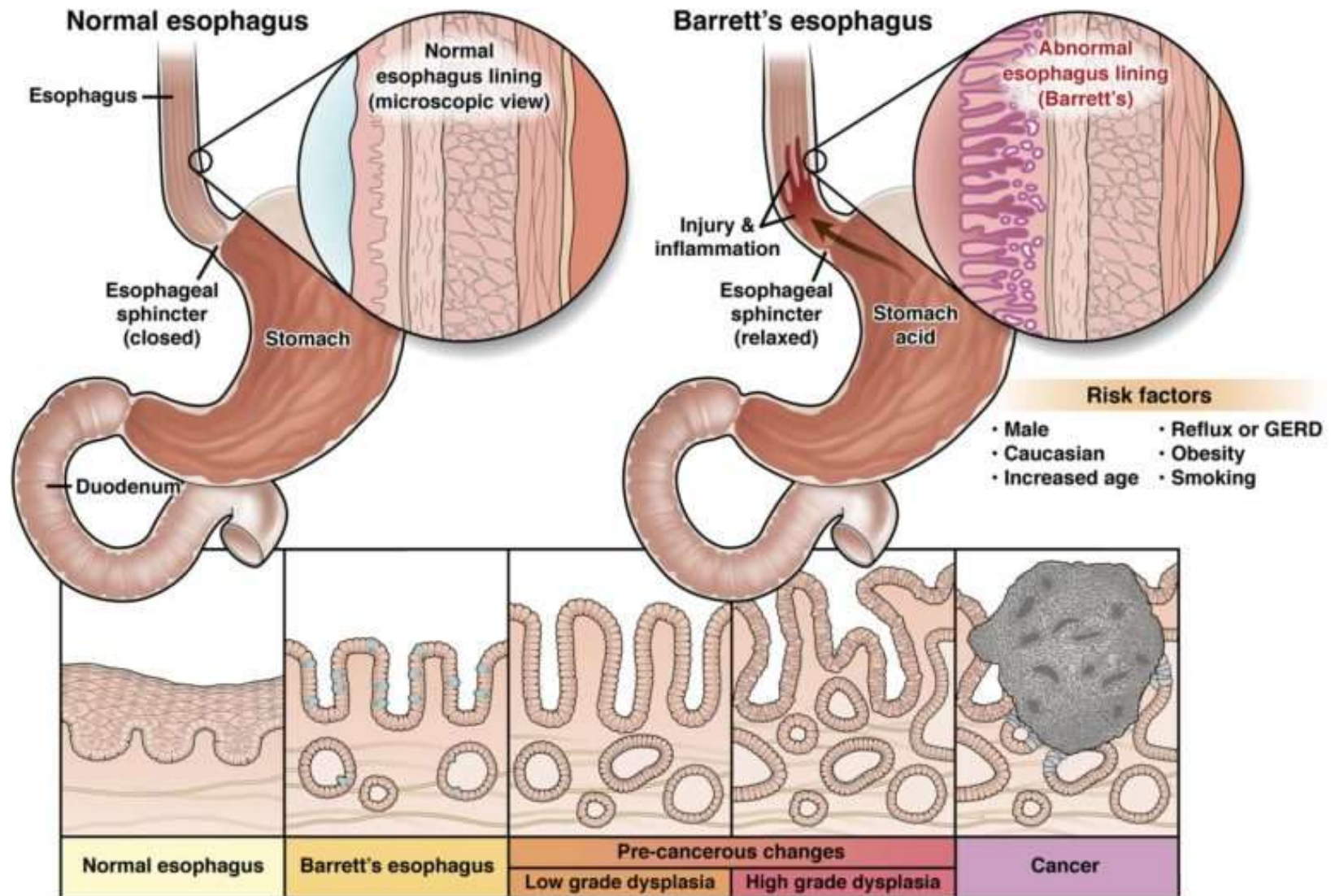
- Dysfunction of the lower esophageal sphincter
- Risk factors
 - Obesity
 - Tobacco usage
 - Genetic disposition
 - Impaired salivary gland secretion
 - Overconsumption of acidic carbonated beverages
 - Hernias

Gastroesophageal Reflux Disease (GERD)

- **Risk of adenocarcinoma of the esophagus**
- Symptoms
 - Typical
 - Heart burn, regurgitation, and dysphasia
 - Atypical
 - Non-cardiac related chest pain, asthma, hoarseness, and pneumonia
- Oral findings
 - Malodor, dysgeusia, tooth erosion

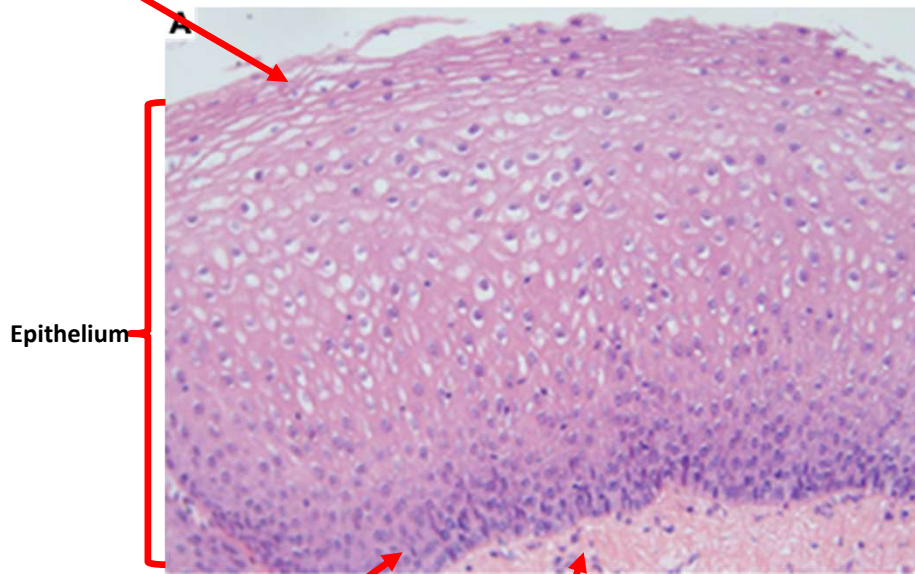


dentagama.com/news/how-acid-reflux-and-gerd-can-damage-your-teeth



Normal Esophagus VS Barret's Esophagus

Superficial layer of epithelium – non-keratinized squamous cells



Columnar cell (tall and rectangular and shape)



Connective tissue-found under the epithelium.

atlasgeneticsoncology.org/solid-tumor/5591/esophagus-barretts-esophagus-dysplasia-and-adenocarcinoma

California Northstate University College of Dental Medicine

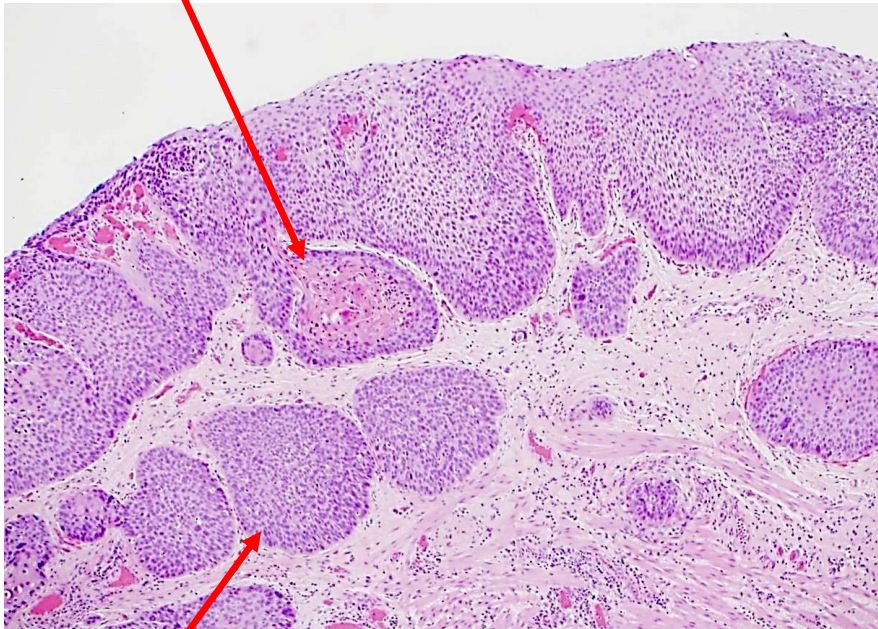
Esophagus - Neoplasms

- Tumors of the esophagus may derive from
 - Squamous epithelium (**squamous cell carcinoma**)
 - Glandular epithelium (**adenocarcinoma**)

Esophageal Cancer

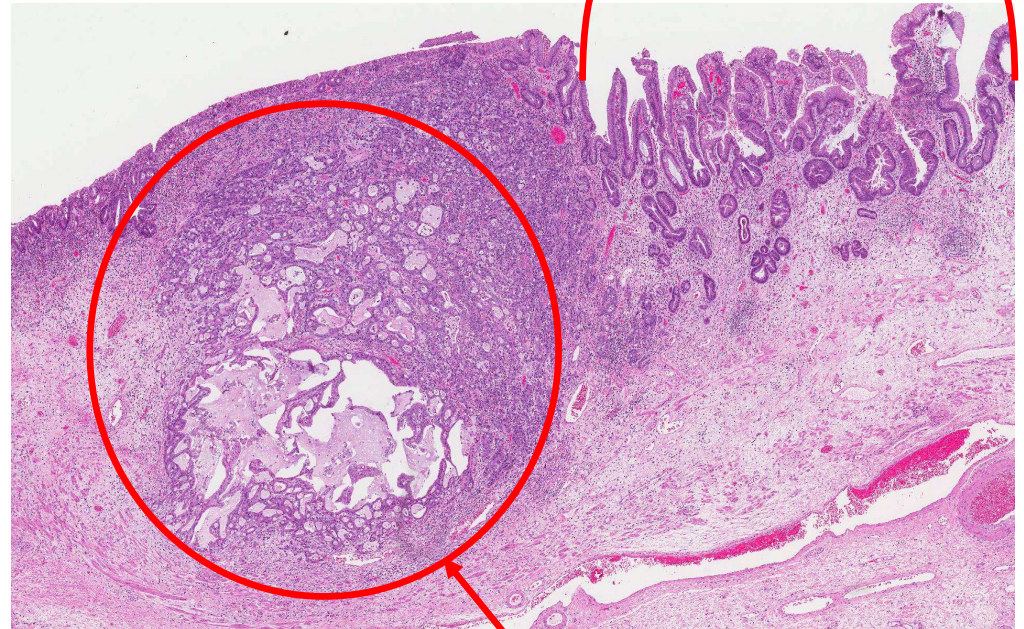
Tumor island separating from the surface epithelium

Squamous cell carcinoma of the esophagus



Tumor island invading the connective tissue= cancer/carcinoma

Adenocarcinoma of the esophagus



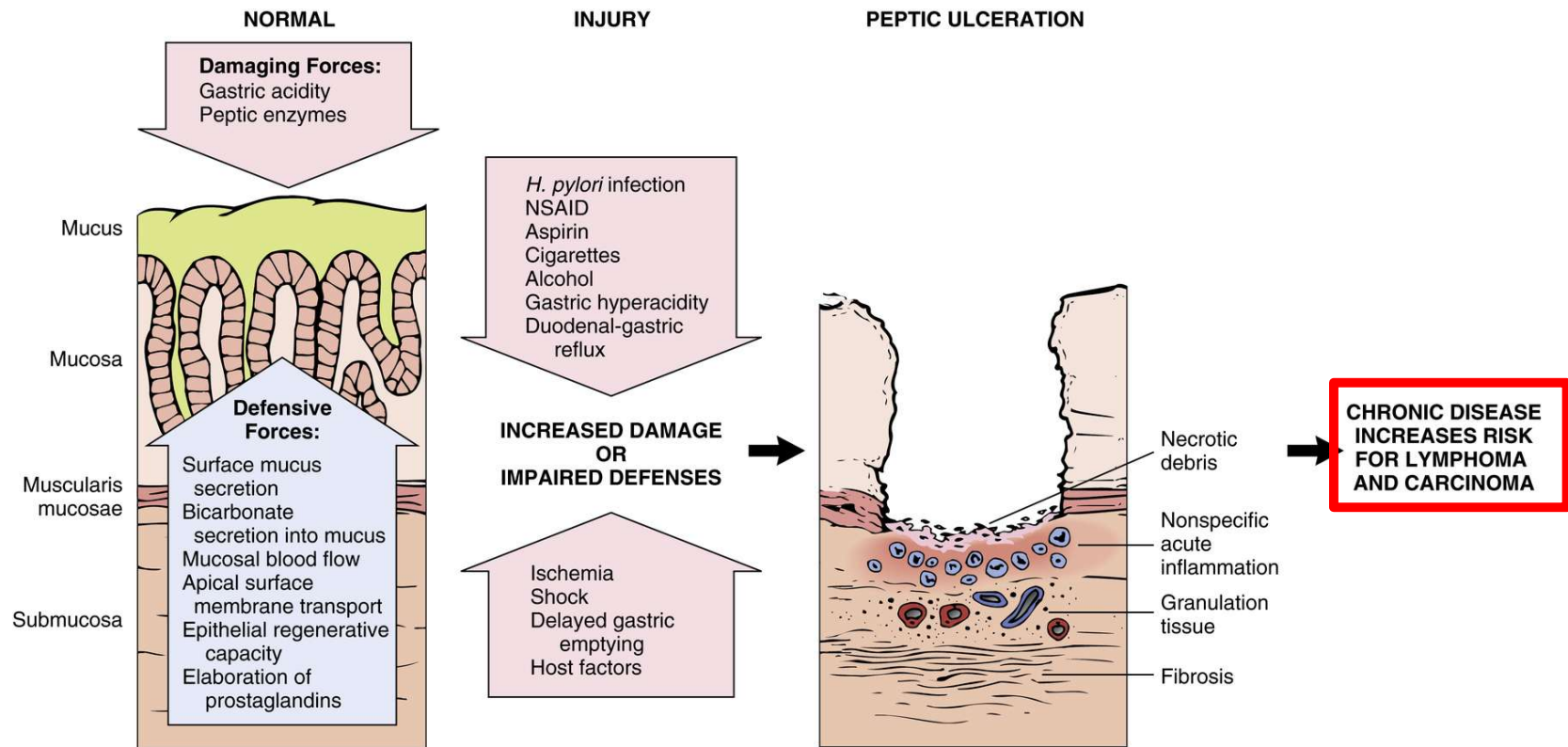
Gastric mucosa

Glandular tumor island invading the connective tissue= cancer/adenocarcinoma

Pathology Outlines

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Peptic Ulcer disease



Little and Falace Dental Management of the Medically Compromised Patient. 10th ed.

Peptic Ulcer Disease

- Signs and symptoms **(Table 11.4)**
 - Asymptomatic
 - Symptomatic
 - Gastric – Symptoms occur after eating, nausea/vomiting, and weight loss
 - Duodenal – Symptoms occur at night and patients have hunger sensations
 - Consuming food, milk, or antacids reduces pain
 - Oral findings
 - Dark, red tongue with a yellow coating
 - Candidiasis
 - Signs of anemia in patients with *H. pylori*-induced ulcers

Pseudomembranous Colitis

- Inflammation of the intestinal lining presenting with abdominal pain, diarrhea, and fever.
- Cause: *Clostridium difficile*
 - Gram positive, anerobic bacteria
 - Produced endotoxins that cause inflammation of the intestinal lining and cellular apoptosis.
- Associated with antibiotic usage: Broad-spectrum antibiotics
 - Clindamycin

Pseudomembranous Colitis



empendium.com/mcmtextbook/image/B31.031_1211.

Celiac Disease

- Prevalent GI disease in the U.S. and Western Europe
- Gluten sensitivity
 - Abnormal reaction to **gliadin**
 - An immunologic inflammatory reaction occurs when wheat products are consumed.
- Malabsorption
 - GI upset: pain and diarrhea
- Hereditary predisposition

Oral Manifestations of Celiac Disease

- Resemble signs of malnutrition
 - Aphthous ulcers
 - Atrophic glossitis
 - Risk of candidiasis
 - Xerostomia
 - Defects to the dentition

Aphthous Ulcer
"Canker sore"



IBS	WHAT'S THE DIFFERENCE?	IBD
IBS = IRRITABLE BOWEL SYNDROME	WHAT DOES IT MEAN?	IBS = INFLAMMATORY BOWEL DISEASE
A group of symptoms that include stomach pain, diarrhea and constipation	WHAT IS IT?	A group of inflammatory diseases including Chron's disease and ulcerative colitis
Twice as common in women as in men	WHO CAN GET IT?	Men and women equally, and it tends to run in families
Usually in late adolescence or early adulthood	WHEN DOES IT HAPPEN?	Usually before you're 30, but it can develop later
Food, stress, medicine, hormones and other factors	WHAT TRIGGERS IT?	There are often no clear triggers
Based on symptoms	HOW IS IT DETECTED?	Usually by colonoscopy, biopsy or other tests
YES	ABDOMINAL PAIN?	YES
YES	CONSTIPATION OR DIARRHEA?	YES
NO	INFLAMMATION OF THE GUT?	YES
NO	ANEMIA?	YES
NO	FEVER?	YES
NO	WEIGHT LOSS?	YES
NO	INCREASED RISK OF CANCER?	YES
Impaired quality of life	COMPLICATIONS	Joints, kidneys, eyes, skin and bones can all be effected
Management through diet, medication and various medical therapies	HOW IS IT TREATED?	Management through diet, medication and various medical therapies

Inflammatory Bowel Diseases

- Ulcerative Colitis
- Crohn's Disease

Ulcerative Colitis

- Affects the colon and can involve one or multiple layers
- Mucosal ulceration is present from rectum to the more proximal portion of the large intestines.
- Clinically: Abdominal pain, bloody diarrhea
- Increased risk of adenocarcinoma
- Oral manifestation
 - Pyostomatitis vegetans



Crohn Disease

- Can involve the entire length of the GI tract.
- All four layer of the colon can be involved.
- “Skip lesions” are seen throughout the colon.
- Clinically: Abdominal pain, bloody diarrhea.
- Increased risk of adenocarcinoma with time.

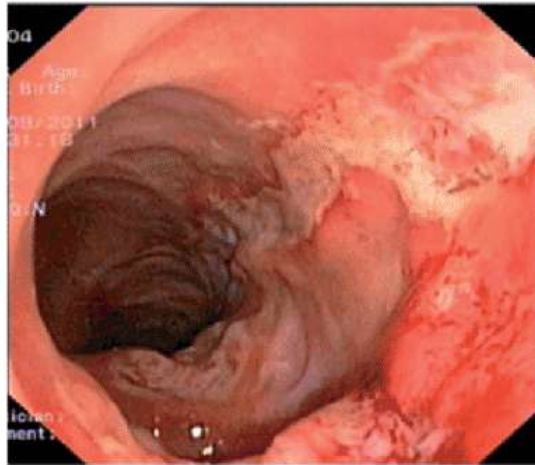
Crohn Disease-Oral Finding



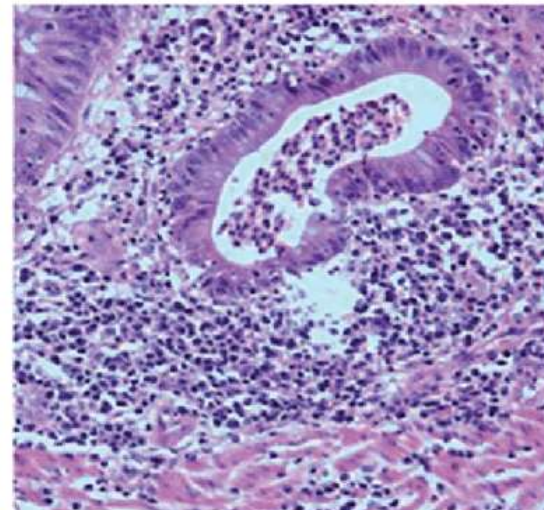
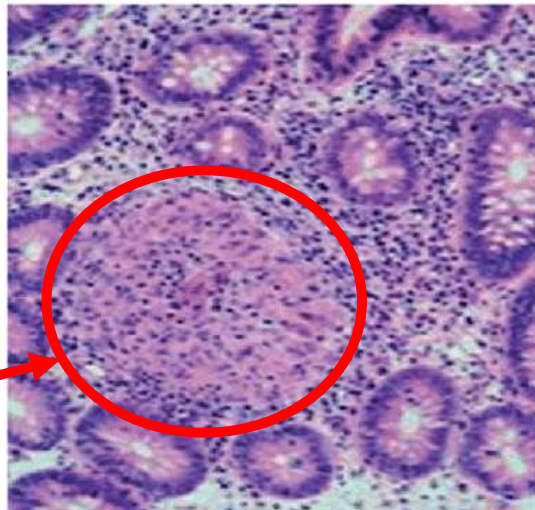
Crohn's disease

Ulcerative colitis

Endoscopy



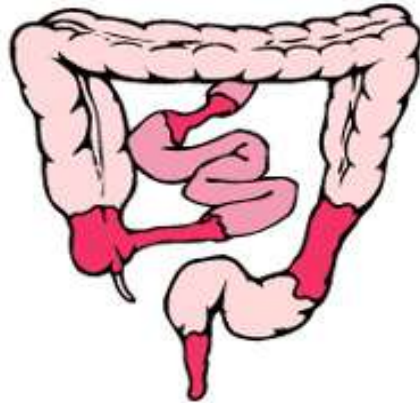
Histology



Granulomatous
Inflammation in
Crohn's Disease

de Bruyn, Magali et al. (2016). The molecular biology of matrix metalloproteinases and tissue inhibitors of metalloproteinases in inflammatory bowel diseases. Critical Reviews in Biochemistry and Molecular Biology. 51. 1-64.

CROHN DISEASE



Skip lesions

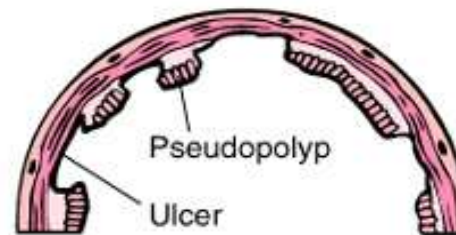
ULCERATIVE COLITIS



Continuous
colonic involvement,
beginning in rectum



Transmural inflammation
Ulcerations
Fissures



Pseudopolyp
Ulcer

Robbin's Basic Pathology 10th Edition Fig. 15.26

Pharmacological Management of IBD

Review Table 11.6

Class of Drugs	Indication	Dental Considerations
Aminosalicylates (5-aminosalicylic acid derivatives)	Ulcerative colitis – distal ileum and colon Small role in Crohn's disease	CBC w/Diff, renal and liver function should be tested Oral inflammation (stomatitis)
Corticosteroids	May be used in combinations with other medications to reduce "flares" Periods of severe disease	Risk of infections, poor wound healing, periodontal disease, and potential adrenal crisis
Immunomodulators	Cases not responding to corticosteroids Weaning patients off steroids Limited use due to toxicity	CBC w/Diff, renal and liver function should be tested Risk of infection and lymphoma Oral inflammation (stomatitis) Gingival hyperplasia (Cyclosporine)
Antibiotics (Metronidazole)	Manage abscesses, maintain remission, and post intestinal surgery	Liver function test Patient should avoid alcohol
Biologics	Severe cases that fail to respond to other medications IV infusion	CBC w/diff, liver function test, and hepatitis B serologies Risk of lymphoma Risk of progressive multifocal leukoencephalopathy

Common IBD Surgical Procedures



Bowel Resection:

Surigical removal of a portion of the small and/or large intestine

Colectomy:

Surigical removal of the entire colon (large intestine)

Proctocolectomy:

Surigical removal of the entire colon (large intestine)

Fistulotomy:

Surigical treatment of anal fistula

Strictureplasty:

Surigical widening of intestinal passageway

Create your own infographic and learn more at:
crohnscolitisfoundation.org/AwarenessWeek

#myIBD

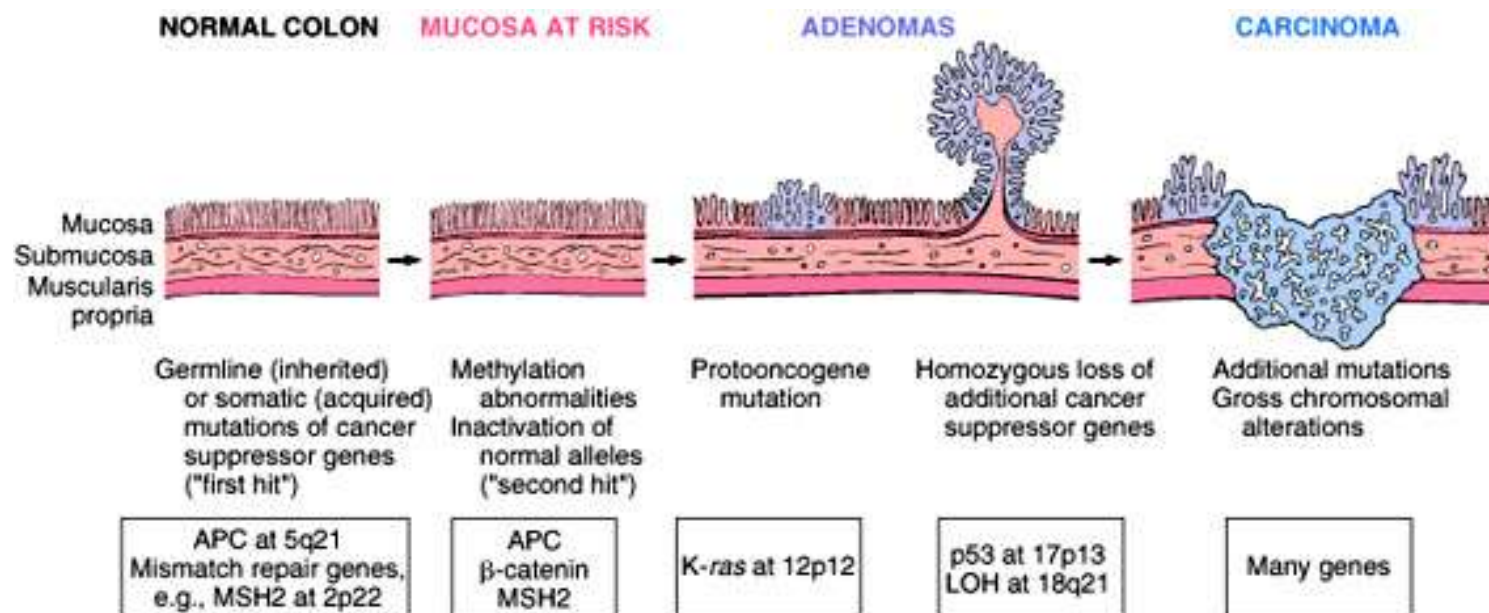
Colonic Diseases

Colon - Polyps

- A protrusion into the lumen
- Many different types
 - Hyperplastic
 - Adenomatous (dysplastic)
- Adenomatous polyps may give rise to adenocarcinoma
 - Screening colonoscopy and polypectomy



www.webpathology.com/image.asp?n=1&Case=221



Robbin's Basic Pathology 10th Edition Fig. 15.26

Neoplasms

- Adenocarcinoma
 - Most common malignancy of the colon
- Carcinoids (rare, except for rectal)
- Other
 - Lymphoma
 - Metastatic carcinoma



Colon – Hereditary Cancer

- Syndromes with mutations which can cause patients to develop cancer at a young age
- **FAP (Familial Adenomatous Polyposis)**
 - **Gardner Syndrome**
- **HNPCC (Hereditary Nonpolyposis Colon Cancer)**
 - **Lynch Syndrome**



Wismayer, Richard. "Familial Adenomatous Polyposis Coli in East Africa: A Case Report and Review of the Literature." *Journal of Advances in Medicine and Medical Research* (2020): 74-80.

Dental Modification for GI Disorders

- Consider chair positioning for patients with GERD and Peptic ulcers.
- Fluoride and products to help with dry mouth.
- Delaying treatment for patients with active oral ulcers and GI upset.
- Providing non-NSAID analgesics and careful selection of antibiotics.
- Prescribing antibiotics which are least likely to be associated with pseudomembranous colitis.
- Decreased response of the immune system and cancer risk in patients with disease pharmacological management.
- Coordination treatment with the medical team in patients undergoing cancer therapy.
- Early screening for patients with a family history of GI cancers

Resources

